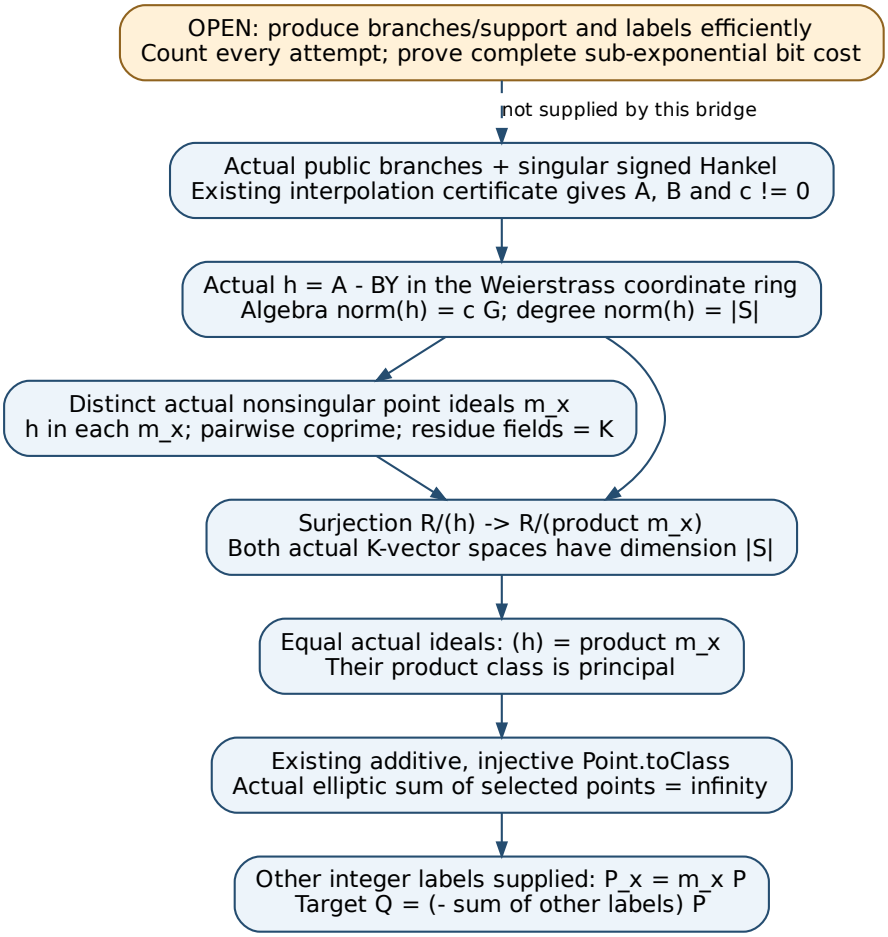
Actual norm certificates to elliptic group relations

2026-11-13. Source`6ba718256`. Unsigned evidence [pcc:unverified].

Let K be a field, W an actual Weierstrass curve with $a_1 = a_3 = 0$, and $f = X^3 + a_2X^2 + a_4X + a_6$. Let S have distinct x coordinates and actual nonsingular points $P_x = (x, z(x))$. For $G = \prod_{x\in S}(X - x)$,

$$A(x) = z(x)B(x), \qquad A^2 - fB^2 = cG, \qquad c \neq 0 \quad \Longrightarrow \quad \sum_{x\in S} P_x = 0.$$

The final public theorem derives the certificate from singularity of the actual barycentric signed Hankel matrix, for $|S| = 2a$, $a \geq 2$. Neither a principal-ideal factorization nor a divisor/group relation is assumed in that theorem [pcc:unverified].



With all other integer labels supplied, $P_i = m_iP$ for $i \neq t$, the actual target satisfies $Q = (-\sum_{i\neq t} m_i)P$. Its label is not a premise. This certificate-to-group bridge does not produce the successful branches or the other labels. See unsigned claims`PCC-20261113-c0da71` and `PCC-20261113-3f21d1` [pcc:unverified].

Actual-ring proof and fixed native controls

Proof. Work in the actual ring $R = K[X,Y]/(W)$ with Mathlib's rank-two $K[X]$ basis $(1, Y)$. For $h = A - BY$, its actual algebra norm is $A^2 - fB^2$. Each polynomial row gives $X - x \mid A - z(x)B$, so $h \in \mathfrak{m}_x = (X - x, Y - z(x))$ by an explicit generator expression. Distinct x coordinates make these actual ideals coprime; their residue fields are K. CRT gives $\dim_K R/(\prod \mathfrak{m}_x) = |S|$. The nonzero principal quotient $R/(h)$ is finite dimensional through its Smith-factor decomposition, and its dimension is $\deg \mathrm{Nm}(h) = |S|$. The quotient map is surjective between equal finite dimensions, hence injective, so $(h) = \prod \mathfrak{m}_x$. Its product class is principal. The existing additive, injective Mathlib Point.toClass yields the actual point sum. Separating the target summand gives the integer scalar relation [pcc:unverified].

This reuses the established Angdinata–Xu ITP2023 ideal-class foundation, sections3.1–3.3, rather than assuming a projective divisor library. Nonsingularity is an explicit premise at every actual point. The field's DecidableEq instance supports the library's point group. Repeated x coordinates are outside the finite-set formulation. The original prime-field objective retains its $p > 3$ domain. No map to the earlier W-jet family is assumed [pcc:unverified].

Frozen controls. The one native artifact uses $p = 17$, $f = X^3 + 2X + 2$ and the following points, fixed prospectively:

$$(0,6), (3,1), (5,1), \quad (6,3), (10,11), (13,7).$$

Control	Recorded result
Positive branches	All on curve, sum infinity, $\det M = 0$
Full norm reconstruction	$A = [14,2,7,3]$, $B = [8,1]$, $c = 9$; norm exactly cG
Flip only $(13,7)$ to $(13,10)$	$\mathrm{Sum}(10,6) = -2(13,7)$; $\det M = 1$, kernel0
Base$(5,1)$, labels$[7,4,1,2,16,8]$	Every supplied label matches by multiplication
Target fixed at node0	Other labels sum31; $(-31)P = (0,6)$
Off-curve$(0,13)$; out-of-range inputs	Reject before group addition

Unsigned native claim`PCC-20261113-476fe0`. Artifact SHA256:

`3875f7bce92f789b273d026b3ea1aabada4eca4c559e92a65c9363f21432e0e5`. JSON null means infinity. No order assertion or scalar search is used. Unit tests reuse these fixtures; they are not independent yield trials [pcc:unverified].

Literal identity certificate. `IDC1hedd7e46a60a6ad5c` seals only the expansion

$$\det \left(\begin{smallmatrix} A & -FB \\ -B & A \end{smallmatrix} \right) = A^2 - FB^2.$$

The unchanged canonical native checker accepts32 statement-derived and16 extra replay points over its v1 fixed modulus. This does not interpret the norm or execute Lean. Canonical hash/checker tests002 pass22; identity clippy001 retains four existing style errors in unchanged canonical SHA256 under Rust1.99.1, without suppression. Separate main native14 tests and strict clippy pass. All evidence remains unsigned [pcc:unverified].

Formal statement map and exact evidence scope

Lean4.35.0.

CryptoLean `148f5dc389b341401e06d76abe7783c4ccf42f2e`.

Mathlib `cfc55fc58714a59dc06e6e79edc6d1d95140d3c2`.

Module build010 and whole-study gate exit0:270 statements,0 additional trusted axioms. Import-inclusive fresh replay session36245 exits0. Each new statement uses only propext, Classical.choice and Quot.sound. Default-PATH PCC checks remain source-only not_covered, distinct from the explicit toolchain checks; claims are unsigned [pcc:unverified].

NormGroupBridge declaration	Inputs and conclusion
quotientFinite	Nonzero actual$h$; finite-dimensional$R/(h)$
idealsEqualOfDimension	Inclusion, both finite equal dimensions; actual ideal equality
pointIdealsCoprime	Distinct$x$; actual point ideals coprime
pointQuotientDimension	Injective$x$, actual nonsingular points; quotient dimension=count
principalPointProduct	Actual membership, nonzero$h$, exact norm degree; principal product
pointSumOfPrincipal	Actual nonsingular points, principal product; actual point sum0
normDegreePointSum	Derives that principal product from membership and norm degree
normFunctionNorm	$a_1 = a_3 = 0$; actual algebra norm of actual$A - BY$
normFunctionAtPoint	Actual$A(x) = yB(x)$; actual point-ideal membership
fullNormPointSum	Rows, nonsingular points, normcG, $c \neq 0$; actual point sum0
cubicDegree	Actual cubic has degree3 for arbitrary lower coefficients
publicHankelPointSum	Exact cardinality, $a \geq 2$, actual matrix determinant0; actual point sum0
targetScalarFromSum	Actual point sum0, other supplied labels; target integer
publicHankelTargetScalar	Composes actual public certificate, point sum and supplied-label scalar

The actual CRT/residue-field linear equivalence pointQuotientEquiv is constructed within the dimension proof. Full normalized statement hashes, source commit and17 unsigned claim records appear in `group-bridge-evidence/claim-pins.json`; raw exits and source pins are in `checks.json`, `fresh-status.json` and`source-inputs.sha256`. Failed prototype001/002, module001–009 and identity attempts are preserved. The identity tests001 include_str failure preceded completion of sparse hydration; after that file was confirmed, tests002 pass without source edits [pcc:unverified].

Sources checked. KB MCP and executable wrapper are unavailable; bounded fallback reads the pinned Mathlib Affine/Point, FreeModule/Norm, IdealQuotient, Ideal operations/quotients and finite-dimensional APIs, plus Angdinata–Xu ITP2023 sections3.1–3.3, `https://doi.org/10.4230/LIPIcs.ITP.2023.6`. Source-review receipt`sources-checked.json`, unsigned claim `PCC-20261113-cee56b`. No novelty or exhaustive-survey claim [pcc:unverified].

Next obligation. Produce actual branches/support and usable labels with a constrained discrete finder, count every attempt and establish complete expected bit cost. The originalexp($o(\log r)$) acceptance criterion is retained. This component supplies neither a visit bound nor a map from the earlier$W$-base jet proposal. No performance graph, benchmark or leaderboard value changes; this study's diagram and current problem record are updated [pcc:unverified].